Cross-Attention for Visual-Inertial Odometry

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1 Introduction

Visual-inertial odometry (VIO) is a fundamental task in robotics and computer vision, enabling the estimation of motion by combining visual and inertial measurements. Transformer-based approaches, such as the VIFT architecture [1], have demonstrated promising results by modeling temporal dependencies through attention mechanisms. However, VIFT relies on a relatively simple fusion strategy, the concatenation of visual and inertial features, which may limit the ability to capture structured cross-modal relationships.

This work modifies the VIFT architecture by implementing a cross-attention-based approach to visual-inertial fusion. The best-performing baseline VIFT model is first trained, and the performance of cross-attention models is then compared against it. The goal of this work is to evaluate whether more structured fusion mechanisms can improve the integration of visual and inertial information for accurate pose estimation.

2 Training Setup

All models were trained on an NVIDIA A100 GPU using the AdamW optimizer with a learning rate of 1×10^{-4} and a batch size of 128. Cosine-annealing warm restarts were used to adjust the learning rate during training. Each model was trained for 200 epochs, with the final evaluation performed using the checkpoint from the best validation epoch. Unless otherwise specified, all architectural modifications used the same training configuration to ensure fair comparison between models.

3 Baseline Model Reproduction

3.1 Evaluation Setup

Following the evaluation protocol described in the VIFT paper [1], model performance was evaluated on KITTI Sequences 05, 07, and 10 from the KITTI odometry dataset [2]. The KITTI odometry dataset consists of real-world driving sequences with synchronized camera and inertial measurements, making it a widely used benchmark for visual-inertial odometry. Model performance was evaluated using the KITTI odometry protocol, which measures drift over trajectory subsequences ranging from 100 m to 800 m. The evaluation metrics are defined as follows:

- **Relative Translation Error (t_{rel})** Measured as the percentage drift in translation.

- **Relative Rotation Error (r_{rel})** Measured in degrees per 100 meters (°/100m), representing rotational drift.

The results are reported separately for each sequence.

3.2 Model Configuration

Model	Parameters (Millions)
VIFT	11.4
Cross-Attention ($d_{\text{model}} = 768$, $d_{\text{ff}} = 128$)	20.9
Cross-Attention ($d_{\text{model}} = 512$, $d_{\text{ff}} = 1024$)	13.3
Cross-Attention ($d_{\text{model}} = 512$, $d_{\text{ff}} = 1536$)	15.4
Cross-Attention ($d_{\text{model}} = 512$, $d_{\text{ff}} = 2048$)	17.5

Table 1: Parameter counts for evaluated model configurations.

The reproduced baseline model was trained using the VIFT repository with the configuration described in the original paper [1]. Key training parameters used for reproduction are summarized in Table 2.

Parameter	**Value**
Sequence length (N)	11
Transformer layers	4
Attention heads	6
Embedding dimension	768
Feed-forward dimension	128
Loss function	L1
Rotation weight (α)	40
Optimizer	AdamW
Learning rate	1×10^{-4}
Batch size	128
Training epochs	200
RPMG	Enabled
Data balancing	Not used

Table 2: Configuration parameters used to reproduce the VIFT baseline model.

Data balancing was not applied during training, as the original paper reported inconsistent performance improvements when this technique was used.

3.3 Baseline Evaluation

The reproduced baseline results are compared with those reported in the original VIFT paper in Table 3. Translation errors (t_{rel}) were generally comparable to the reported values, while rotational errors (r_{rel}) were generally higher across all sequences. This discrepancy suggests that rotational estimation may be more sensitive to training dynamics or subtle implementation details that were not explicitly documented in the original paper or repository.

Model	Seq. 05		Seq. 07		Seq. 10	
	$t_{rel}(\%)$	$r_{rel}(°/100\text{m})$	$t_{rel}(\%)$	$r_{rel}(°/100\text{m})$	$t_{rel}(\%)$	$r_{rel}(°/100\text{m})$
VIFT (Reported)	**2.02**	**0.53**	1.75	**0.47**	2.11	**0.39**
VIFT (Reproduced)	2.29	0.85	**1.61**	1.11	**1.94**	0.69

Table 3: Relative translation and rotation error for the reported and reproduced VIFT baselines on KITTI Sequences 05, 07, and 10.

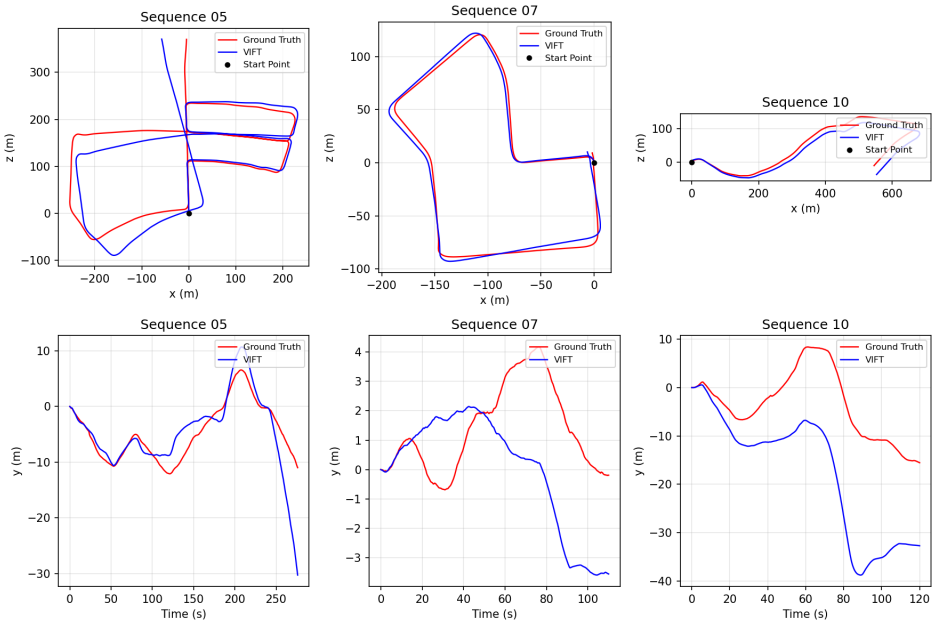

Figure 1: Baseline VIFT predicted and ground-truth trajectories on KITTI Sequences 05, 07, and 10. Top row: top-down trajectory comparison in the x–z plane. Bottom row: vertical position (y) over time.

Figure 1 illustrates the predicted and ground-truth trajectories for KITTI Sequences 05, 07, and 10. Motion in the x–z plane represents the top-down trajectory, while vertical motion is shown separately as y displacement over time. Overall, the reproduced model demonstrates similar large-scale trajectory behavior, indicating that the overall trajectory structure is successfully captured. However, noticeable divergence occurs in the vertical (y) direction over time. These results establish a reliable baseline for evaluating the cross-attention-based models explored in subsequent experiments.

4 Architecture Modifications

4.1 Cross-Attention Baseline

Cross-Attention ($d_{\text{model}} = 768$, $d_{\text{ff}} = 128$) was trained using the same hyperparameters as the reproduced VIFT configuration, with the standard PoseTransformer module replaced by a cross-attention transformer architecture. In this design, the IMU latent first queries the visual representation through cross-attention, followed by causal temporal self-attention over

the IMU latent sequence.

Model	Seq. 05		Seq. 07		Seq. 10	
	$t_{rel}(\%)$	$r_{rel}(°/100\,\mathrm{m})$	$t_{rel}(\%)$	$r_{rel}(°/100\,\mathrm{m})$	$t_{rel}(\%)$	$r_{rel}(°/100\,\mathrm{m})$
VIFT (Reported)	**2.02**	**0.53**	1.75	**0.47**	2.11	**0.39**
VIFT (Reproduced)	2.29	0.85	**1.61**	1.11	**1.94**	0.69
Cross-Attention ($d_{\text{model}} = 768$, $d_{\text{ff}} = 128$)	2.64	1.00	1.99	0.68	2.14	0.85

Table 4: Relative translation and rotation error for the reported VIFT baseline, reproduced VIFT baseline, and Cross-Attention ($d_{\text{model}} = 768$, $d_{\text{ff}} = 128$) on KITTI Sequences 05, 07, and 10.

Table 4 compares Cross-Attention ($d_{\text{model}} = 768$, $d_{\text{ff}} = 128$) with the reported and reproduced VIFT models. This configuration exhibits higher translation error across all sequences compared to the other models. However, rotational performance remains competitive, particularly on Sequence 07, where rotational error is reduced relative to the reproduced VIFT model. This outcome was expected, as the current dimensional configuration is not well matched to the requirements of cross-attention. Cross-attention involves projecting modality representations into a shared embedding space where compatible query, key, and value representations can be computed. Here, both the visual and IMU latents were projected into a shared embedding space of dimension $d_{\text{model}} = 768$, in order to remain consistent with the VIFT architecture.

However, this choice introduces unnecessary computational overhead. The original latent dimensions are $d_{\text{visual}} = 512$ and $d_{\text{IMU}} = 256$, suggesting that projecting both modalities into a shared embedding space of $d_{\text{model}} = 512$ is a reasonable choice. This dimensional configuration preserves the full visual representation while expanding the IMU representation without requiring compression of either modality. Reducing the embedding dimension also decreases the number of parameters in the projection layers and reduces the overall computational cost of the model. This configuration is evaluated in the following experiment.

4.2 Cross-Attention with Modified Dimensions

Model	Seq. 05		Seq. 07		Seq. 10	
	$t_{rel}(\%)$	$r_{rel}(°/100\,\mathrm{m})$	$t_{rel}(\%)$	$r_{rel}(°/100\,\mathrm{m})$	$t_{rel}(\%)$	$r_{rel}(°/100\,\mathrm{m})$
VIFT (Reported)	**2.02**	**0.53**	1.75	**0.47**	2.11	**0.39**
VIFT (Reproduced)	2.29	0.85	**1.61**	1.11	**1.94**	0.69
Cross-Attention ($d_{\text{model}} = 768$, $d_{\text{ff}} = 128$)	2.64	1.00	1.99	0.68	2.14	0.85
Cross-Attention ($d_{\text{model}} = 512$, $d_{\text{ff}} = 1024$)	2.17	0.60	3.01	0.69	2.30	0.49

Table 5: Relative translation and rotation error for the reported VIFT baseline, reproduced VIFT baseline, and Cross-Attention models on KITTI Sequences 05, 07, and 10.

As shown in Table 1, Cross-Attention ($d_{\text{model}} = 512$, $d_{\text{ff}} = 1024$) reduces parameter count by approximately 36% relative to Cross-Attention ($d_{\text{model}} = 768$, $d_{\text{ff}} = 128$). The initial cross-attention configuration used an embedding dimension of $d_{\text{model}} = 768$ with a feed-forward network dimension of $d_{\text{ff}} = 128$. Although this modified configuration reduced parameter count and enabled faster training, such a narrow feed-forward network may have introduced a representational bottleneck that limited performance. Feed-forward layers positioned between attention blocks are responsible for transforming latent representations

and introducing features learned during training that are not explicitly present in the current input sequence.

In the modified configuration, the embedding dimension was reduced from $d_{\text{model}} = 768$ to $d_{\text{model}} = 512$, motivated by the latent dimensionalities of the individual modalities ($d_{\text{visual}} = 512$, $d_{\text{IMU}} = 256$). This configuration preserves the full visual representation while expanding the IMU representation without requiring compression of either modality. The feed-forward dimension was increased to $d_{\text{ff}} = 1024$, consistent with common transformer design practice where feed-forward layers are wider than the embedding dimension [3].

The previous configuration used $h = 6$ attention heads. After reducing the embedding dimension to $d_{\text{model}} = 512$, the number of attention heads was increased to $h = 8$ such that

$$d_{\text{model}} \bmod h = 512 \bmod 8 = 0.$$

This ensures that the embedding dimension can be evenly divided across attention heads, allowing uniform partitioning of feature channels across attention heads.

4.2.1 Effect of Rotational Loss Weight

Table 5 shows that although Cross-Attention ($d_{\text{model}} = 512$, $d_{\text{ff}} = 1024$) significantly reduced rotational error relative to both Cross-Attention ($d_{\text{model}} = 768$, $d_{\text{ff}} = 128$) and the reproduced VIFT baseline on Sequence 05, translation errors increased substantially on Sequences 07 and 10 compared to the other models. This behavior may be influenced by the design of the loss function used in the VIFT architecture.

$$\begin{aligned} \mathcal{L}_r &= \|R - R_{\text{gt}}\|, \\ \mathcal{L}_t &= \|t - t_{\text{gt}}\|, \\ \mathcal{L} &= \mathcal{L}_t + \alpha \mathcal{L}_r \end{aligned} \tag{1}$$

Here, $\mathcal{L}_r$ represents rotational error and $\mathcal{L}_t$ represents translational error. The parameter α is a weighting factor used to balance the contributions of rotational and translational loss. The VIFT paper reports that $\alpha = 40$ provided the best performance. However, the architectural modifications introduced in this work may have altered the relative magnitudes and optimization dynamics of $\mathcal{L}_t$ and $\mathcal{L}_r$, necessitating re-tuning of α.

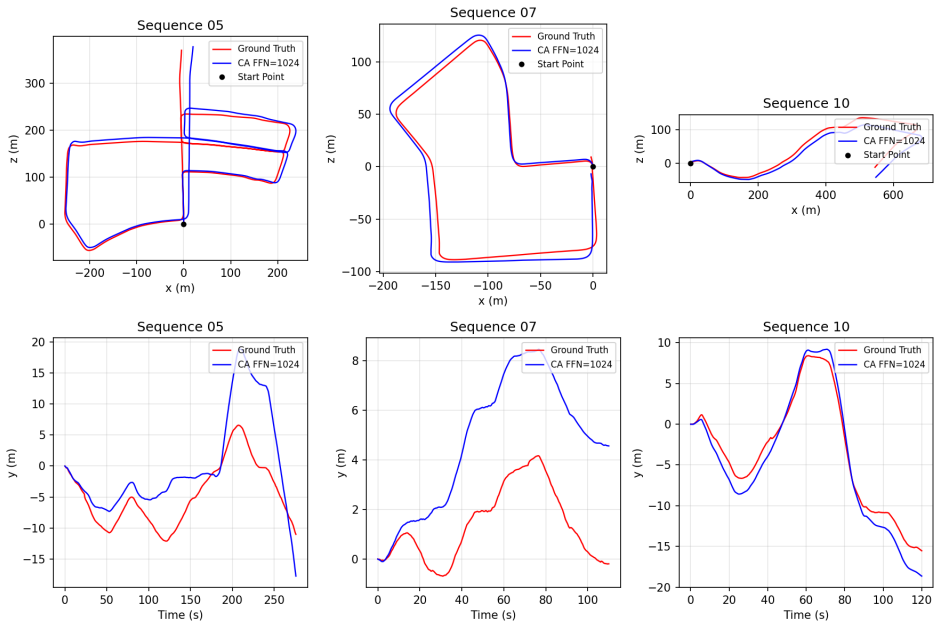

Figure 2: Cross-Attention ($d_{\text{model}} = 512$, $d_{\text{ff}} = 1024$) predicted and ground-truth trajectories on KITTI Sequences 05, 07, and 10.

A larger value of α places greater emphasis on rotational accuracy during optimization, which may inadvertently reduce the relative contribution of translational error minimization. This imbalance may contribute to the large deviations from the ground-truth y-axis trajectories observed in Sequences 05 and 07, as shown in Figure 2. A sweep over values of α and the resulting model performance are summarized in Table 6.

	Seq. 05		Seq. 07		Seq. 10	
α	$t_{rel}(\%)$	$r_{rel}(°/100\text{m})$	$t_{rel}(\%)$	$r_{rel}(°/100\text{m})$	$t_{rel}(\%)$	$r_{rel}(°/100\text{m})$
VIFT (Reported)	2.02	0.53	1.75	0.47	2.11	0.39
VIFT (Reproduced)	2.29	0.85	**1.61**	1.11	1.94	0.69
10	2.76	0.98	3.57	2.36	2.98	0.98
20	2.39	0.66	3.13	1.47	2.67	0.87
23	2.45	0.68	2.25	0.46	**1.90**	0.58
25	1.99	0.55	1.76	0.73	1.93	0.43
27	**1.81**	0.57	2.18	0.53	2.03	0.51
30	2.03	0.50	2.12	**0.38**	2.29	0.47
35	2.11	**0.49**	2.41	0.41	2.02	**0.36**
40	2.17	0.60	3.01	0.69	2.30	0.49

Table 6: Effect of the rotational weighting factor α on relative translation and rotation error across KITTI Sequences 05, 07, and 10, with VIFT baselines included for reference.

As shown in Table 6, $\alpha = 25$ provided the most balanced cross-attention performance across sequences, yielding low translation and rotational error without introducing large degradation on any individual sequence. Lower values of α generally degraded both translation and rotation performance, likely because poorer rotational estimates also

propagated into translation error. Larger values of α tended to emphasize rotational accuracy, although this did not consistently improve translation performance.

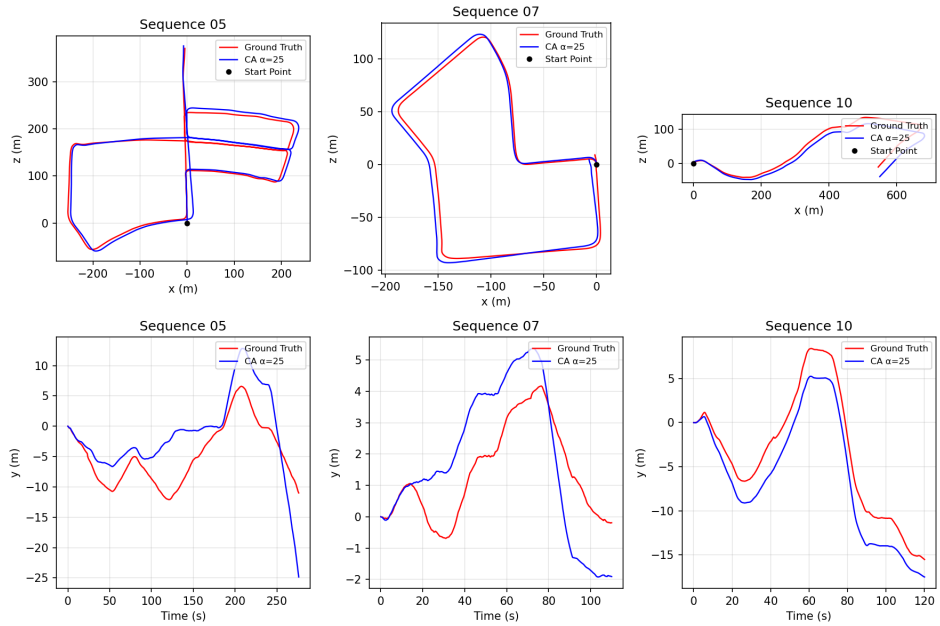

Figure 3: Cross-Attention ($d_{\text{model}} = 512$, $d_{\text{ff}} = 1024$) predicted and ground-truth trajectories after tuning the rotational loss weight to $\alpha = 25$ on KITTI Sequences 05, 07, and 10.

Figure 3 shows a clear reduction in trajectory overshoot along the y-axis compared to the behavior observed in Figure 2. This result indicates that the optimal rotational loss weight is architecture-dependent and should be re-tuned whenever significant modifications are made to the model structure.

4.2.2 Effect of Feed-Forward Network Dimension

	Seq. 05		Seq. 07		Seq. 10	
d_{ff}	$t_{rel}(\%)$	$r_{rel}(°/100\text{m})$	$t_{rel}(\%)$	$r_{rel}(°/100\text{m})$	$t_{rel}(\%)$	$r_{rel}(°/100\text{m})$
VIFT (Reported)	2.02	**0.53**	1.75	0.47	2.11	**0.39**
VIFT (Reproduced)	2.29	0.85	**1.61**	1.11	1.94	0.69
1024	1.99	0.55	1.76	0.73	**1.93**	0.43
1536	**1.88**	0.55	2.00	**0.43**	2.24	0.53
2048	2.31	0.69	2.30	0.74	2.21	0.60

Table 7: Effect of feed-forward network (FFN) dimension on relative translation and rotation error across KITTI Sequences 05, 07, and 10, with VIFT baselines included for reference.

Table 7 summarizes the effect of increasing the FFN dimension while keeping $\alpha = 25$ fixed. Although the results for larger FFN dimensions are comparable to those obtained with $d_{\text{ff}} = 1024$, no consistent performance improvement is observed as model capacity increases. Notably, increasing the FFN dimension to 2048 resulted in degraded performance

across all sequences, suggesting that excessive capacity may lead to overfitting. The absence of consistent performance improvements despite increased model capacity suggests that the model is not limited by representational capacity. Additionally, substantial variation in performance across sequences indicates that the learned representations may not generalize consistently across different motion patterns. These observations suggest that introducing regularization may help prevent overfitting and improve generalization. One straightforward approach is to apply dropout within the feed-forward layers of each transformer block, which can reduce reliance on specific activations and encourage more robust feature learning. Although modifying d_{ff} may necessitate re-tuning of α, the next experiment instead investigates the effect of dropout as a regularization strategy.

4.2.3 Effect of Dropout

	Seq. 05		Seq. 07		Seq. 10	
Dropout	$t_{rel}(\%)$	$r_{rel}(°/100\text{m})$	$t_{rel}(\%)$	$r_{rel}(°/100\text{m})$	$t_{rel}(\%)$	$r_{rel}(°/100\text{m})$
VIFT (Reported)	2.02	0.53	1.75	**0.47**	2.11	**0.39**
VIFT (Reproduced)	2.29	0.85	**1.61**	1.11	1.94	0.69
0.00	1.99	0.55	1.76	0.73	1.93	0.43
0.05	**1.81**	0.57	1.77	0.87	1.89	0.56
0.10	1.96	**0.49**	2.00	0.56	**1.81**	0.48

Table 8: Effect of feed-forward network dropout rate on relative translation and rotation error across KITTI Sequences 05, 07, and 10, with VIFT baselines included for reference.

Table 8 summarizes the results of applying dropout to the FFN layers of the Cross-Attention model with $d_{\text{model}} = 512$, $d_{\text{ff}} = 1024$, and $\alpha = 25$. Similar to the results observed when increasing the FFN dimension, the effect of dropout is not consistent across sequences. These results suggest that moderate dropout can improve performance on selected sequences, but does not provide a uniform generalization benefit.

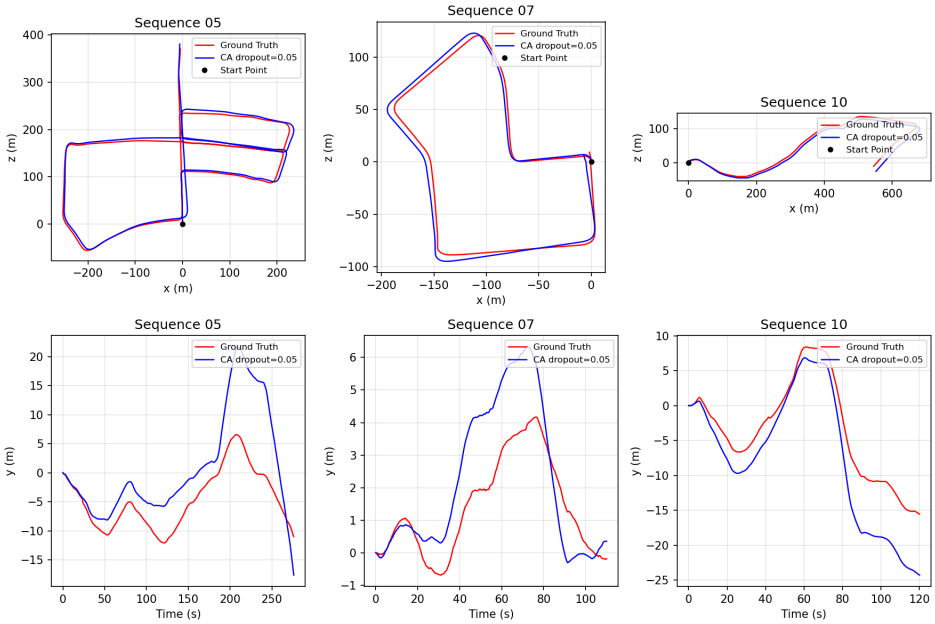

Figure 4: Cross-Attention ($d_{\text{model}} = 512$, $d_{\text{ff}} = 1024$, $\alpha = 25$, dropout $= 0.05$) predicted and ground-truth trajectories on KITTI Sequences 05, 07, and 10.

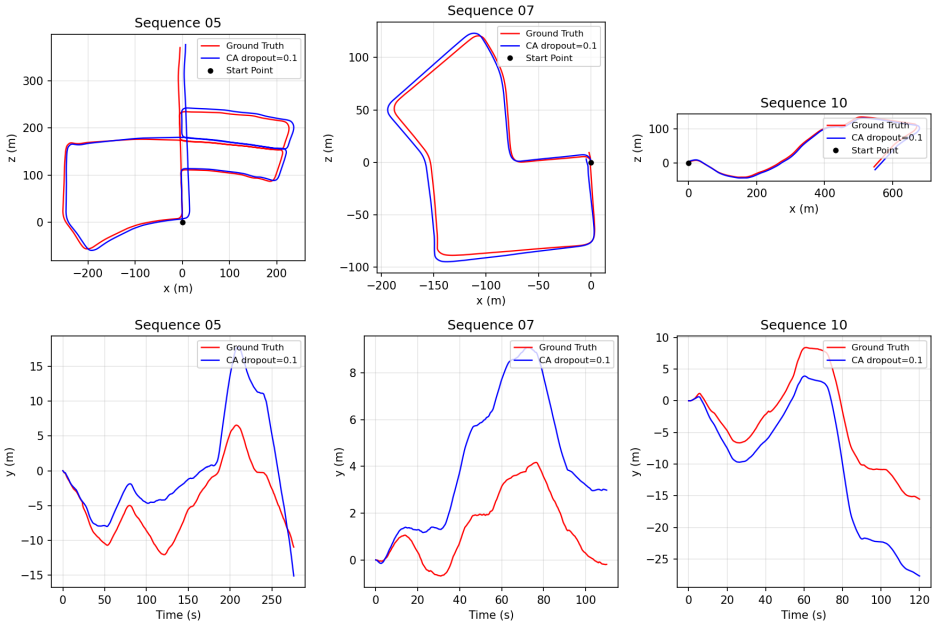

Figure 5: Cross-Attention ($d_{\text{model}} = 512$, $d_{\text{ff}} = 1024$, $\alpha = 25$, dropout $= 0.10$) predicted and ground-truth trajectories on KITTI Sequences 05, 07, and 10.

Figures 4 and 5 show substantial vertical trajectory overshoot, particularly on Sequences 05 and 07. This behavior is similar to that observed in Figure 2 prior to tuning α. However, inspection of the relative contributions of the translational and weighted rotational

components of the loss indicated that they were already balanced. Because the reproduced VIFT model does not exhibit similarly aggressive y-axis spikes, and because visual features in VIFT are explicitly concatenated into the latent representation passed through the model, it is hypothesized that visual features may be underutilized in the cross-attention configuration. In contrast to concatenation-based fusion, this cross-attention design incorporates visual information only through queries from the IMU latent, which may weaken the influence of visual features during fusion. Additionally, since IMU data are sampled at a higher frequency and often contain strong motion cues, the model may learn to rely more heavily on IMU latents, further reducing the contribution of visual information. This possible modality imbalance may be especially problematic for vertical pose estimation, where visual cues may play an important stabilizing role. To investigate this hypothesis, the next experiment evaluates an IMU-only ablation.

4.2.4 IMU-Only Ablation

To evaluate the contribution of visual features, an IMU-only ablation was conducted. Accordingly, the cross-attention from IMU latents to visual latents was removed, and only causal temporal self-attention over IMU latents was preserved. All other components of the model architecture remained unchanged.

	Seq. 05		Seq. 07		Seq. 10	
Model	$t_{rel}(\%)$	$r_{rel}(°/100\,\mathrm{m})$	$t_{rel}(\%)$	$r_{rel}(°/100\,\mathrm{m})$	$t_{rel}(\%)$	$r_{rel}(°/100\,\mathrm{m})$
CA ($d_{\mathrm{model}} = 512$, $d_{\mathrm{ff}} = 1024$, $\alpha = 25$)	1.99	0.55	1.76	0.73	1.93	0.43
IMU-Only	7.38	0.69	13.39	0.67	6.93	0.73

Table 9: Relative translation and rotation error for Cross-Attention ($d_{\mathrm{model}} = 512$, $d_{\mathrm{ff}} = 1024$, $\alpha = 25$) and the IMU-only ablation on KITTI Sequences 05, 07, and 10.

Table 9 shows that the IMU-only model produces substantially higher translation error across all evaluated sequences compared to Cross-Attention ($d_{\mathrm{model}} = 512$, $d_{\mathrm{ff}} = 1024$, $\alpha = 25$). This result indicates that visual features contribute meaningfully to pose estimation performance.

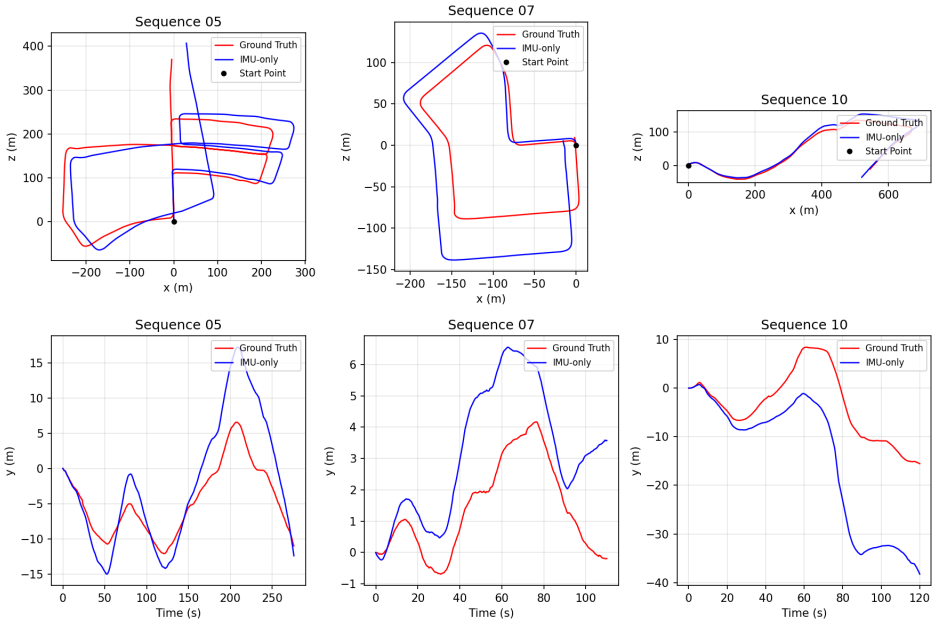

Figure 6: IMU-only predicted and ground-truth trajectories on KITTI Sequences 05, 07, and 10. Top row: top-down trajectory comparison in the x–z plane. Bottom row: vertical position (y) over time.

However, Figure 6 shows vertical (y) trajectory instability similar to that observed in the cross-attention models after applying dropout. This observation suggests that although visual information is utilized in this configuration, the current fusion mechanism may not fully leverage visual cues to stabilize vertical motion, for which visual features may play a more significant role in correcting drift.

4.3 Summary of Additional Experiments

Beyond the primary experiments described above, several additional architectural modifications were explored, including increasing model depth, applying the data balancing strategy described in the VIFT paper, and introducing residual visual skip connections to reinforce visual information flow. However, none of these modifications produced consistent performance improvements across sequences. Similarly, increasing the penalty applied to the y-axis error did not produce measurable improvements in vertical trajectory stability. Additionally, initial experiments using Kendall uncertainty loss [4] to learn the α weight produced worse results than the optimal $\alpha = 25$ obtained from the manual sweep. These findings suggest that performance improvements are more strongly influenced by fusion strategy than model capacity or loss re-weighting alone.

5 Discussion

Across the evaluated configurations, the cross-attention models achieved performance that was competitive with the reproduced VIFT baseline on several sequence-level metrics. Modifications to the embedding dimension and feed-forward network capacity produced

measurable improvements in rotational accuracy, particularly on Sequences 05 and 10. The final cross-attention configuration used a reduced embedding dimension of $d_{\text{model}} = 512$, an expanded feed-forward network dimension of $d_{\text{ff}} = 1024$, and a rotational loss weight of $\alpha = 25$. Additional regularization through dropout improved performance on selected sequences, but not consistently across all sequences, and was therefore not included in the final configuration.

Despite these gains, vertical trajectory estimation remained a persistent challenge across cross-attention configurations. Although top-down (x–z) motion was accurately captured, instability along the y-axis suggests that current fusion mechanisms may not sufficiently integrate visual information for vertical pose correction. The IMU-only ablation confirmed that visual information contributes meaningfully to overall performance, but that visual features may not be fused in a manner that effectively stabilizes vertical motion.

6 Conclusion

This work evaluated cross-attention as an alternative to the concatenation-based fusion used in the VIFT architecture. The reproduced VIFT baseline provided a reliable reference point for comparing architectural modifications, and the final cross-attention configuration improved selected sequence-level error metrics while remaining competitive with the reproduced baseline. Overall, cross-attention is a viable fusion strategy for visual-inertial odometry, but robust multimodal fusion remains the primary challenge. Further improvements are most likely to come from better fusion design.

7 Future Work

Several directions remain for improving multimodal fusion performance in transformer-based visual-inertial odometry. The most important is the development of more effective fusion strategies that encourage balanced use of visual and inertial features. Although cross-attention was explored in this work, persistent instability along the y-axis suggests that vertical pose estimation remains challenging under the current fusion design. Promising alternatives include bidirectional cross-attention, joint attention mechanisms, and modality-specific constraints or loss formulations designed to better stabilize vertical motion. More comprehensive modality ablation studies may also provide deeper insight into the relative contributions of visual and inertial information.

Another direction is improving the tuning of the rotational loss weight α. Kendall uncertainty loss may provide a more principled way to learn an appropriate value of α, reducing the need for extensive manual tuning during model comparison. A targeted manual sweep could then be used to further refine the most promising configurations.

Finally, additional regularization experiments may improve generalization. Although FFN dropout improved performance in selected cases, other strategies such as weight decay, modality dropout, or dropout applied in other parts of the attention blocks may be worth exploring. Reducing the batch size may also improve generalization by increasing stochasticity in the gradient updates.

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References

[1] Y. B. Kurt, A. Akman, and A. A. Alatan, "Causal transformer for fusion and pose estimation in deep visual inertial odometry," 2024.

[2] A. Geiger, P. Lenz, and R. Urtasun, "Are we ready for autonomous driving? the kitti vision benchmark suite," in *2012 IEEE Conference on Computer Vision and Pattern Recognition*, pp. 3354–3361, 2012.

[3] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," 2023.

[4] A. Kendall, Y. Gal, and R. Cipolla, "Multi-task learning using uncertainty to weigh losses for scene geometry and semantics," 2018.